µSquad

USER MANUAL

ANALOG FUNCTION GENERATOR

Thank you for choosing the µSquad Function Generator. This device is a versatile tool designed for generating fundamental electronic waveforms for various applications.

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1. INTRODUCTION

Primary Fuction
To generate essential waveforms: sine, square, triangular, and sawtooth.

Design
It is an Analogue Function Generator with a robust, self-contained power supply.

Key Parameters
All four waveforms feature adjustable parameters for Frequency, Amplitude, and DC Offset Voltage.

Special Feature
The duty cycle can be precisely adjusted when generating a square wave.

2. SAFETY INFORMATION

Power Supply
The device accepts standard AC wall power and converts it internally. **Do not attempt to apply DC power directly.** The internal circuitry operates using a +/- 12V dual-supply, and the final signal output range is from +12V to -12V.

Environment
Do not expose the device to moisture, rain, or liquids. Avoid placing it near heat sources (amplifiers, radiators, sunlight).

Ventilation:
Ensure ventilation grills remain unobstructed

Servicing
Disconnect power before servicing.

3. INSTALLATION & SETUP

STEP 01
Connect AC Power Input
Connect a standard AC power cord into the power socket on the device. Then, plug the cord into a standard wall outlet. The device's internal circuitry safely handles the conversion from the AC wall power to the necessary +/- 12V DC supply.

STEP 02
Connect Output to Test Equipment
Connect your test equipment (e.g., oscilloscope) using the dedicated signal terminals. Connect the oscilloscope probe's hook clip to the Ground terminal and the probe tip to the Signal Output terminal. The final signal output range is from +12V to -12V.

4. PRODUCT OVERVIEW : FRONT PANEL CONTROLS

A. WAVEFORM & FREQUENCY SELECTION SWITCHES

These switches determine the basic shape and the coarse frequency range of the output signal.

Switch	Name	Function	Detailed Operation
Switch 1	Waveform Type Selector	Selects the primary waveform type for the output signal.	Turn to the desired position to output Sine, Square, Triangle, or Sawtooth.
Switch 2	Frequency Range Selector	Selects the coarse frequency range.	Adjust this switch to select the appropriate frequency range before fine-tuning with the knobs. This dictates the range for both Sine and the other waveforms.

B. POTENTIOMETERS (KNOBS)

These five knobs provide fine control over the frequency, shape, amplitude, and DC bias of the signal.

Knob	Control Name	Function	Detailed Operation
P1	Sine Frequency Deviation	Provides fine frequency adjustment for the Sine wave within the range set by Switch 2.	Use this knob to precisely set the Sine wave frequency after selecting the coarse range.
P2	Square/Triangle Frequency Deviation	Provides fine frequency adjustment for Square, Triangle, and Sawtooth waves within the range set by Switch 2.	This knob works similarly to P1 but affects the other three waveforms.
P3	PWM / Duty Cycle	Adjusts the Pulse Width Modulation (PWM) or the Duty Cycle of the Square wave.	Rotate to change the symmetry/pulse width of the square wave. (This control is non-functional for Sine and Triangle waves).
P4	Amplitude	Controls the peak-to-peak voltage of the output signal.	Rotate clockwise to increase the output amplitude. The maximum output is +/- 12V.
P5	DC Offset	Controls the DC bias applied to the waveform.	Rotate to shift the entire signal waveform above or below 0V (the offset).

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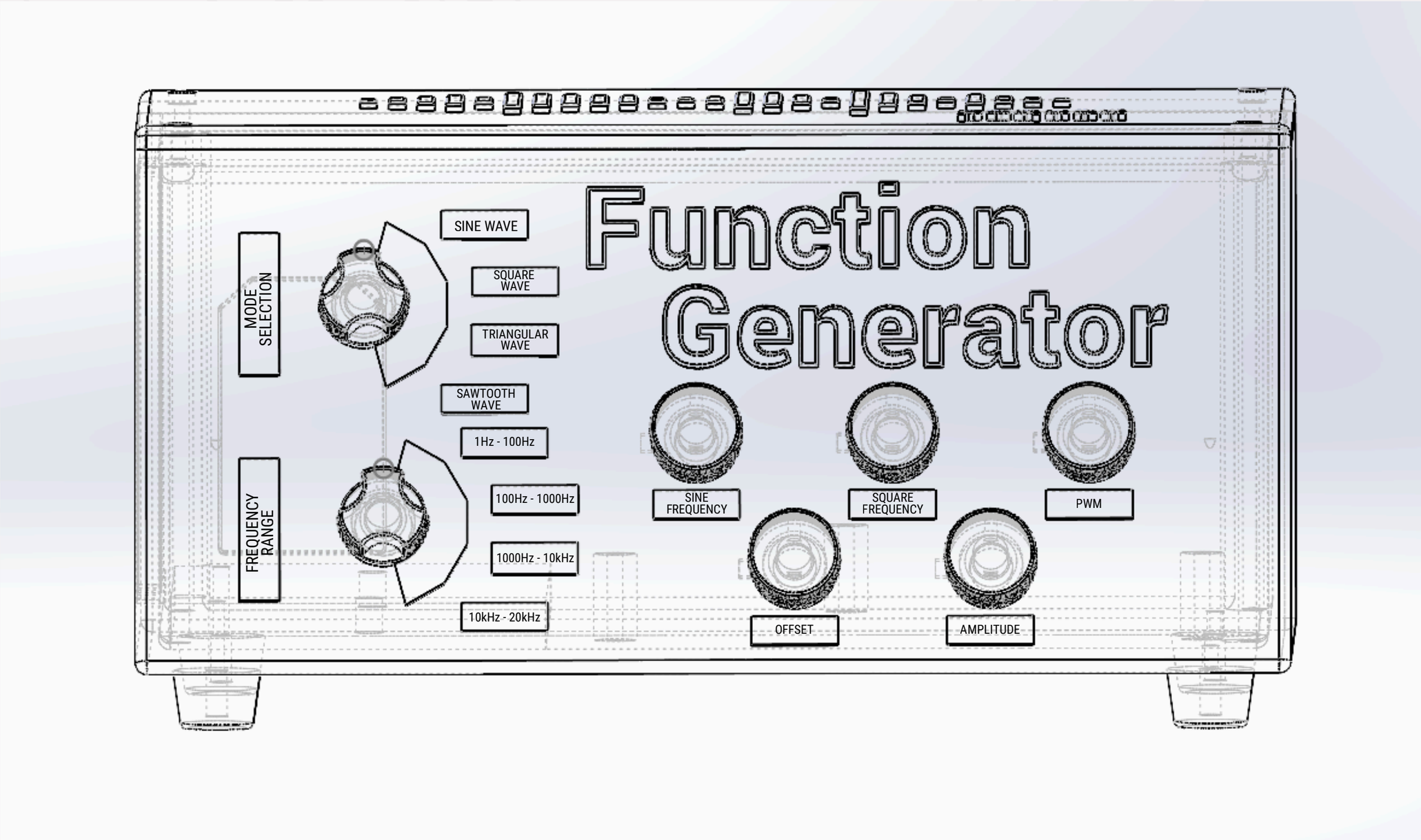

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